Kali Linux

What Is Kali Linux?

- **Kali Linux** is a **specialized operating system** built for **cyber-security, ethical hacking, and digital forensics**.
- It's based on **Debian Linux** and comes pre-loaded with **hundreds of security tools** used by penetration testers, researchers, and security students.
- **Main Purpose of Kali Linux**
- To provide a **ready-made environment** for **ethical hacking**, **penetration testing**, and **security auditing** — all in one system.

Popular Tools in Kali Linux

Category	Example Tools	Purpose
Information Gathering	`nmap`, `whois`, `dnsenum`	Network scanning and reconnaissance.
Vulnerability Analysis	`nikto`, `OpenVAS`	Scan web servers for known vulnerabilities.
Exploitation	`Metasploit`, `BeEF`	Exploit system weaknesses for testing.
Web Application Testing	`Burp Suite`, `OWASP ZAP`, `sqlmap`	Detect XSS, SQLi, CSRF, etc.
Password Attacks	`John the Ripper`, `Hydra`, `Hashcat`	Test password strength and brute-force.
Wireless Attacks	`aircrack-ng`, `reaver`	Wi-Fi network penetration testing.
Sniffing & Spoofing	`Wireshark`, `Ettercap`	Capture and analyze packets.
Forensics	`Autopsy`, `Binwalk`, `Volatility`	Analyze images, memory dumps, etc.

Who Uses Kali Linux

Role	Use Case
Cybersecurity Students	Learn ethical hacking safely.
Penetration Testers	Test systems for weaknesses.
Network Engineers	Monitor traffic and troubleshoot issues.
Digital Forensic Experts	Investigate cybercrime evidence.

Why We Use Kali Linux in Cybersecurity

Reason	Description	Real-Time Example
🧰 **Pre-Installed Tools**	600 + security tools come built-in (no setup hassle).	Nmap for scanning, Wireshark for packet capture, Burp Suite for web testing.
💻 **Penetration Testing**	Used to test the security of systems, networks, and web apps.	Simulate attacks safely in lab to find vulnerabilities before hackers do.
🔍 **Digital Forensics**	Recover or analyze deleted or damaged data.	Investigate USB or disk images after an attack.
⚙️ **Network Security Analysis**	Monitor and analyze live network packets.	Use Wireshark and tcpdump to find suspicious traffic.
👨‍🏫 **Learning & Training**	Perfect OS for cybersecurity education.	Students can practice ethical hacking without installing tools manually.
🧭 **Customizable & Open Source**	Anyone can modify or add tools.	You can add your own scripts or configurations easily.

Real-Time Scenarios

Scenario	How Kali Linux Helps
1️⃣ Company wants to check if its website is vulnerable.	Use `nmap` to find open ports, `nikto` to test web vulnerabilities.
2️⃣ Suspicious data transfer in a network.	Use `Wireshark` to capture and inspect traffic.
3️⃣ Forgot admin password (authorized recovery).	Use `John the Ripper` to test password hashes.
4️⃣ Investigating cyberattack on a USB drive.	Use `Autopsy` to analyze the drive's contents.

Installation of Kali Linux — Step-by-Step

- **Option 1: Install in VirtualBox (Recommended for Beginners)**
- **Download the ISO**
  - Go to: https://www.kali.org/get-kali/
  - Download **Kali Linux 64-bit Installer ISO**.
- **Install VirtualBox**
  - Download and install from https://www.virtualbox.org/.
- **Create a New Virtual Machine**
  - Name: `Kali Linux`
  - Type: **Linux**
  - Version: **Debian (64-bit)**
  - Memory: **4 GB (4096 MB)** minimum
  - Disk size: **25 GB+**
- **Attach the ISO**
  - Go to **Settings → Storage → Empty CD Drive → Choose a disk file** → select downloaded Kali ISO.
- **Start the VM**
  - Click **Start** → Kali Linux setup begins.
- **Follow the Installer**
  - Choose *Graphical Install*
  - Select language, location, keyboard.
  - Set hostname (e.g., `kali`), username, and password.
  - Partition disk → choose *Guided – use entire disk*.
  - Continue → installation completes → reboot.
- **Login**
  - Username: the one you created
  - Password: (your password)
  - You'll see the Kali desktop (GNOME or XFCE).

After Installation — First Steps:

- Update system

sudo apt update && sudo apt upgrade -y

- Check available tools:

sudo apt list --installed | grep kali

What is sudo in Linux?

sudo stands for “SuperUser DO”.

It allows a regular (non-root) user to run specific commands with administrator (root) privileges without logging in as the root user.

In other words:
sudo gives temporary admin power to a normal user for a single command.

Why do we use sudo?

Because:

- Not every user should have **root (admin)** access all the time.
- Running everything as **root** is risky — one wrong command can break the system.
- With sudo, you can **run only the commands you need with admin rights** — safely and traceably.

Aspect	Normal Command	sudo Command
Privilege Level	Runs as a normal user	Runs as root (administrator)
System Access	Limited to user's home & permissions	Full access to all system files
Effect	Safe — cannot harm system	Risky — can modify or delete system files
Password Needed	No	Yes (user's password for sudo authentication)
Use Case	For daily operations	For administrative tasks
Example	`mkdir myfolder`	`sudo mkdir /etc/myfolder`
Can install software?	❌ No	✅ Yes
Can change configurations?	❌ No	✅ Yes